



interfaceNode - *Manual*

Version 1.0.5

Hardware Revision: B.0.x

The Atkin Engineering **interfaceNode** is a Wireless RS-485 bridge that provides bridging connections between RS-485, USB, TCP sockets, and WebSockets for various control devices.

front

rear



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1. Connections

Front

- USB Type-C, USB 2.0 full-speed port [[^]usbpower]
- 1x Status LED
- 1x System button

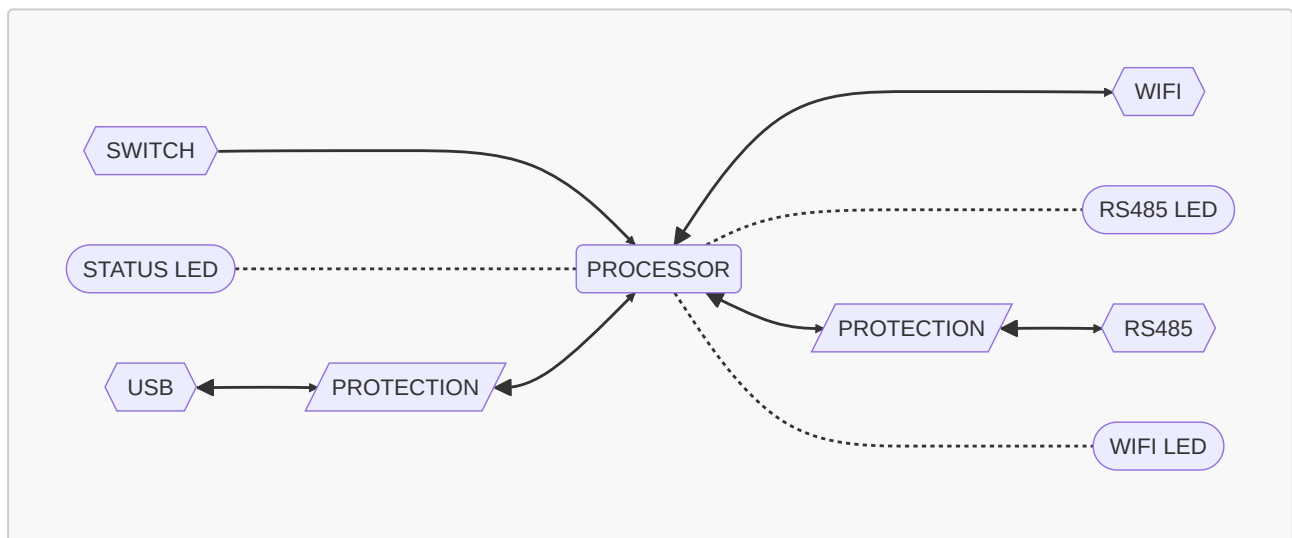
Rear

- 2x RS-485 (parallel RJ12 6P6 ports) supplies power & data [[^]rs485power]
- 2x RGB status LEDs
- Full / Half duplex (hardware jumper - see [[^]jumpers])
- Termination **None** / **120ohm** (hardware jumper - see [[^]jumpers])

RS-485 pinout (RJ12 6P6)

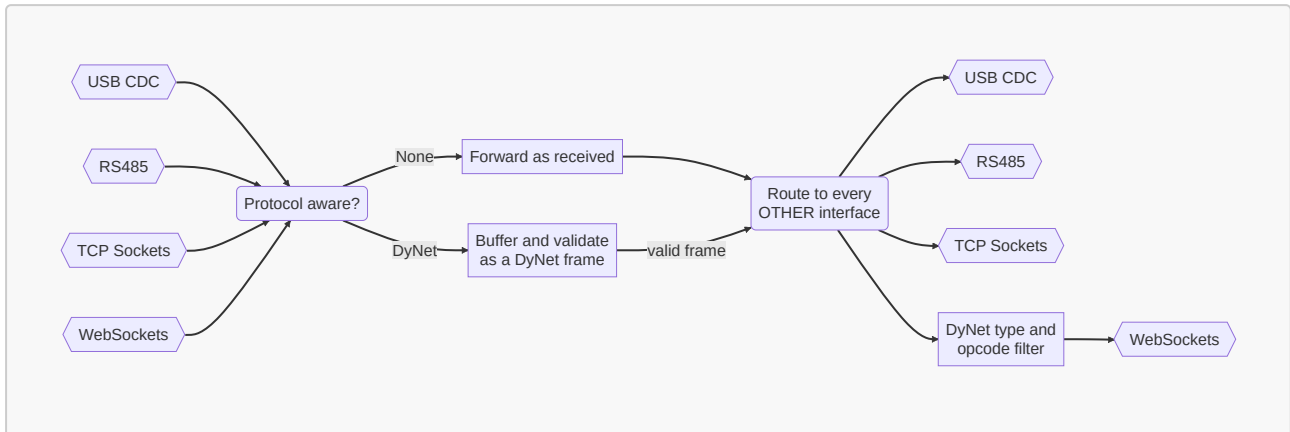
1. Ground
2. Ground
3. Data +
4. Data -
5. 12/24 VDC
6. 12/24 VDC

2. Block diagram



3. Operation flow

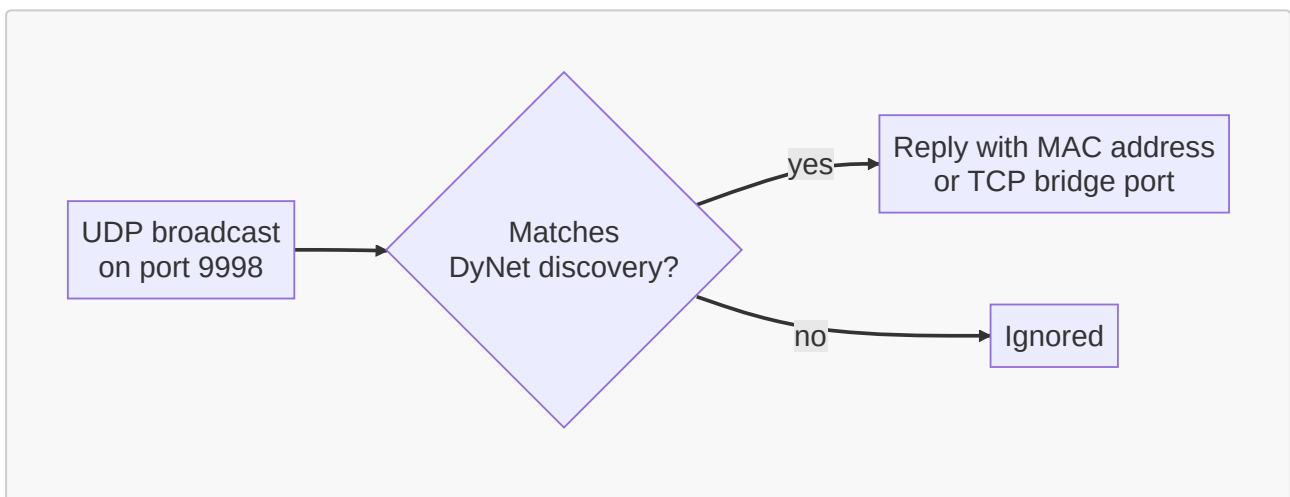
Data arriving on USB, RS485, a TCP socket or a WebSocket is handled the same way, whichever interface it came in on:



A frame is never sent back out of the interface it arrived on. With **Protocol aware** set to **DyNet**, incomplete data is held until the rest of the frame arrives, and data that fails its checksum is discarded while the stream re-synchronises on the next valid frame. With it set to **None**, data is passed straight through with no validation.

Only WebSocket delivery is filtered - see [WebSocket DyNet filtering](#). USB, RS485 and TCP always receive every frame.

The DyNet discovery beacon (see [Sockets](#)) is a separate process. It does not pass through the flow above, and nothing it receives or sends is routed to any bridge interface:



4. Default mode

A factory-reset **interfaceNode** boots in **Access point** mode with:

- SSID: **interfaceNodeXXYY**, where **XXYY** is the device's serial number (the last two bytes of its MAC address, in hex - see [Device](#))
- Password: the device's own default password - see [Default password](#)
- Security: **WPA2/WPA3 PSK**
- IP address: **192.168.4.1**, with a DHCP server for connected devices
- **Protocol aware**: **DyNet**
- TCP port: **50000**, TLS on, login required (see [TCP bridge security](#))
- Web interface over HTTPS; WebSocket bridge over TLS (**wss://**), login required (see [Security](#))
- RS485 baud rate: **9600**
- WebSocket DyNet filtering: DyNet 1 forwarded with opcodes **0-255**; DyNet 2 not forwarded (see [WebSocket DyNet filtering](#))

Note: settings are also restored to these defaults after every firmware update - see [Firmware](#).

Default password

Every **interfaceNode** has its own default password, used for the default access point, for [configuration mode](#), and with the username **admin** for the web interface, WebSocket and TCP logins (see [Security](#)). There is no shared factory password.

- 12 characters, lower-case hex (**0-9**, **a-f**), for example **3fa91c07d2e8**.
- Generated by the device from its MAC address using HMAC-SHA256 and a secret key held by Atkin Engineering. It cannot be worked out from the MAC address, serial number or network name alone.
- Not changed by a factory reset - the device always returns to the same default password.
- Supplied with the device. If it has been lost, contact Atkin Engineering with the device's MAC address (shown under [Device](#) in the web interface, or on the label).

Note: changing the access point password in the web interface replaces the default password for operating mode only. Configuration mode always uses the default password, so the device can still be reached for setup.

5. System button

The system button supports two gestures:

- **Short press** Switches between operating mode and configuration mode: the device reboots into configuration mode if it is operating, or back into operating mode if it is already in configuration mode. See [System](#) for what configuration mode means.
- **2 second hold** Full factory reset - erases all user data and reboots into [default mode](#). See [Factory reset](#).

Note: the button is not active immediately at power-on - it responds once the rest of the system (network, bridge or web interface, beacon) has started. It also stops responding once a reboot has been triggered by any means (the button itself, or an action in the web interface); a press during the short delay before the restart is ignored.

6. Config mode

Configuration mode runs the web interface used to view and change the device's settings. See [System](#) for how it relates to normal bridge operation, and [Default mode](#) / [System button](#) for how to enter and leave it.

In configuration mode the device runs its own access point, `interfaceNodeXXYY`, secured with its [default password](#). Join that network, then open the device in a web browser at its IP address (`https://192.168.4.1` in [default mode](#)), or by its mDNS hostname, e.g. `https://interfacenode3080.local` for serial number `3080`. mDNS works on most networks and operating systems without any setup. Typing the address without `https://` also works: plain `http://` (port 80) redirects to `https://`.

The web interface is served over HTTPS (TLS) using the device's own self-signed certificate (see [Device certificate](#)). Because the certificate isn't issued by a public authority, the browser shows a warning the first time ("Your connection is not private", "Warning: Potential Security Risk Ahead" or similar). Check that the SHA-256 fingerprint the browser shows for the certificate matches the one printed in the system log at start-up or shown under [Security](#) (once you've seen it there over a trusted connection), then choose to continue (for example [Advanced](#) > [Proceed](#)). Most browsers then remember the choice for that address. With HTTPS turned off (see [Security](#)) the interface is plain `http://` on port 80.

The web interface asks for a username and password first. The default is `admin` with the device's [default password](#). After 5 wrong passwords in a row, logging in is refused for 30 seconds, doubling with each further wrong password up to 10 minutes. A login lasts until [Log out](#), 15 minutes without use, or the device restarting.

The page is divided into the sections below. Only the settings that apply are shown: the access point or station settings for the selected WiFi mode, and the DyNet filters only when [DyNet](#) is selected. Hardware-only settings (RS485 termination and duplex - see [Connections](#)) are not part of the web interface.

Device

Read-only details: status, serial number, hostname, MAC address, firmware version, build date, SDK version and the running firmware partition.

Network

Setting	Values
WiFi mode	Access point / Station / No WiFi - see Operation mode
Access point Network name	Text
Access point Password	Text, 8+ characters
Access point Channel	1-13
Access point Security	WPA PSK, WPA2 PSK, WPA/WPA2 PSK, WPA3 PSK, WPA2/WPA3 PSK
Access point Device IP address	Default (192.168.4.1) / Custom (IP address, network mask, gateway, DNS)
Station Network name	Text
Station Security	Open, WEP, WPA PSK, WPA2 PSK, WPA/WPA2 PSK, WPA3 PSK, WPA2/WPA3 PSK, WPA2 Enterprise, WPA3 Enterprise
Station Password	Text, 8+ characters
Station Enterprise sign-in	Username, identity, server certificate validation and certificate upload (Enterprise security only)
Station Device IP address	Automatic (DHCP) (default) / Static (IP address, network mask, gateway, DNS)

Protocol

Setting	Values
Protocol awareness	None / DyNet - see Operation flow
DyNet 1 opcodes	List of opcodes sent to WebSocket clients, e.g. 0-255 - see WebSocket DyNet filtering
DyNet 2 opcodes	As above, for DyNet 2

Bridge

Setting	Values
TCP port	49152-65535 - see Sockets
RS-485 baud rate	9600 / 19200 / 38400 / 57600 / 115200 - see Serial

Security

Setting	Values
Web interface Username	Text, 1-31 characters, no : (default admin)
Web interface New password	Text, 8-63 characters, entered twice. Leave blank to keep the current password
Use TLS for	Web interface, WebSocket bridge, TCP bridge - each on (default) or off. See Encryption (TLS)
WebSocket clients	Login required (default) / Open (no login) - see WebSocket bridge
WebSocket Username	Text, 1-31 characters, no : (default admin)
WebSocket New password	Text, 8-63 characters, entered twice. Leave blank to keep the current password
TCP clients	Login required (default) / Open (no login) - see TCP bridge security
TCP Username	Text, 1-31 characters, no : (default admin)
TCP New password	Text, 8-63 characters, entered twice. Leave blank to keep the current password
Web pages from other sites	Allow (default) or not - cross-origin requests (CORS) to the web interface and the WebSocket /auth URL. Needed by web apps served from elsewhere, such as dynet_ui. Logins are still required either way
Security event log	Record in the system log on (default) or off - see Security events
Device certificate	Shows the certificate in use; upload your own, or make a new self-signed one - see Device certificate

Passwords are never shown or sent back by the device - password fields (including the WiFi passwords under [Network](#)) are always empty, and a password only changes when a new one is typed. The device stores only a salted hash of each login password (PBKDF2-HMAC-SHA256), never the password itself.

Note: choosing [Open \(no login\)](#) lets anyone on the network send and receive DyNet traffic over that bridge. Use it only on a network you trust, for clients that can't log in.

Encryption (TLS)

With TLS on (the default for all three), traffic is encrypted with TLS 1.2 (ECDHE key exchange, AES-GCM), so passwords and DyNet traffic can't be read or changed by others on the network:

Service	TLS on	TLS off
Web interface	<a href="https://<device>/(port 443); http:// redirects to it">https://<device>/(port 443); http:// redirects to it	<a href="http://<device>/(port 80)">http://<device>/(port 80)
WebSocket bridge	<a href="wss://<device>:8080/ws">wss://<device>:8080/ws , login at <a href="https://<device>:8080/auth">https://<device>:8080/auth	ws:// and http:// , same port
TCP bridge	TLS on the TCP port	Plain TCP on the TCP port

Clients must use the matching form: a plain client can't connect to a TLS service, and the reverse. Turning TLS off is a deliberate choice for clients that can't use it, and leaves that traffic - including passwords - readable by anyone on the network. A warning is shown on the page while any service has TLS off. Changes take effect when the settings are saved (the device restarts).

If the web interface can't start HTTPS (for example if no certificate could be made), it falls back to plain HTTP so the device can still be set up, and logs an error. The WebSocket and TCP bridges don't start without a certificate while their TLS is on.

Device certificate

On its first start (and after a [factory reset](#)) each **interfaceNode** makes its own TLS certificate and private key: ECDSA P-256, self-signed, named after the device, with [interfacenodeXXYY](#) and [interfacenodeXXYY.local](#) as alternative names, valid until 2049. Its subject reads [CN=interfacenodeXXYY, serialNumber=XXYY, OU=interfaceNode, O=Atkin Engineering, L=Melbourne, ST=Victoria, C=AU](#), so the maker and device are visible in the browser's certificate viewer (as the certificate is self-signed, this is information only - it doesn't make the browser trust it). The key is made on the device, never leaves it, and is kept in encrypted storage. The certificate's SHA-256 fingerprint is shown here and printed in the system log at start-up.

- [Upload certificate and key](#) Use your own certificate (PEM, optionally with its chain) and its unencrypted private key (PEM), for example one issued by your organisation's certificate authority so browsers trust it without a warning. Both files must be under 8 KB; the device checks that the key matches the certificate before storing it. ECDSA and RSA keys are accepted.
- [Make new self-signed certificate](#) Replaces the certificate and key with new ones, for example if the key may have been exposed.

Either takes effect after the next restart. Browsers and clients that had accepted the old certificate will warn again.

Saving and maintenance

Action	Effect
Save and reboot	Stores every setting and reboots into operating mode
Reboot	Restarts the device without changing any settings
Factory reset	Erases all user data and reboots with factory defaults - see Factory reset
Install firmware	Updates the firmware - see Firmware

See [System](#) for details.

USB drive

In configuration mode the USB Type-C port works as a USB drive instead of a serial port. Connect it to a PC and a 5 MB drive named [interfaceNode storage](#) appears, which can be read and written like a USB stick.

- [device_info.txt](#) Written by the device each time it starts in configuration mode: serial number, MAC address, hostname, mDNS name, WiFi name, firmware version and build, and the running firmware partition. It never contains passwords. Changes made to it are overwritten at the next start.
- Other files are kept until deleted, including across firmware updates. Settings are stored separately and are never visible on the drive.

Note: eject the drive on the PC before saving settings, rebooting or pressing the system button, so files being written are not lost. Do not store passwords or other secrets on the drive - anyone with the device and a USB cable can read it.

7. Operation mode

interfaceNode can operate in three WiFi configurations:

1. **Access point** Creates a WiFi network for other devices to connect to.
2. **Station** Connects to an existing WiFi network as a client.
3. **No WiFi** WiFi is switched off. The bridge runs between USB and RS485 only; the TCP bridge, WebSocket bridge, DyNet discovery beacon and mDNS are not started. Configuration mode still uses its access point, so the device can always be set up.

The **Protocol awareness** setting controls how the bridge handles data:

- **None** Data is forwarded between interfaces as received, with no validation.
- **DyNet** Data is buffered and validated as DyNet 1 or DyNet 2 frames (checksum-checked). Invalid data is discarded and the stream re-synchronises automatically.

Access point

- **Network name** SSID of the access point.
- **Password** Password for the access point.
Note: must be at least 8 characters. A shorter password is not accepted and the device's **default password** is used instead.
- **Channel** WiFi channel (1-13).
- **Security** Security type: **WPA PSK**, **WPA2 PSK**, **WPA/WPA2 PSK**, **WPA3 PSK** or **WPA2/WPA3 PSK**. **WPA2/WPA3 PSK** (the default) is recommended. **Open** and **WEP** are not available for the access point; if either is found in saved settings, the device switches to **WPA2/WPA3 PSK**.

IP address settings:

- **Default** **192.168.4.1**, with the built-in DHCP server.
- **Custom** Set your own IP address, network mask, gateway and DNS server. Leaving gateway or DNS blank offers none to connected devices.

Note: configuration mode always uses the default address, so the device can always be reached for setup even if a custom address is wrong.

Station

- **Network name** SSID of the network to join.
- **Security** Security type of the network being joined: **Open**, **WEP**, **WPA PSK**, **WPA2 PSK**, **WPA/WPA2 PSK**, **WPA3 PSK**, **WPA2/WPA3 PSK**, **WPA2 Enterprise** or **WPA3 Enterprise**.
- **Password** Password of the network being joined. Must be at least 8 characters. For Enterprise networks this is the sign-in password.

If **WPA2 Enterprise** or **WPA3 Enterprise** is selected, an **Enterprise sign-in (EAP)** group appears:

- **Username** EAP username.
- **Identity** EAP identity string.
- **Validate server certificate / Don't validate** Whether to check the RADIUS server's certificate.
- **RADIUS / CA certificate** File picker and **Upload certificate** button. Upload the RADIUS server's CA certificate (PEM format) to enable validation.
Note: the certificate is used from the *next* WiFi connection attempt. If validation is selected but no certificate has been uploaded, the connection proceeds without validation.

PEAP, TTLS and EAP-FAST are supported; the RADIUS server selects the method.

If the device can't join the network, it tries 5 times, waiting 5, 10, 20 and 40 seconds between attempts (about two minutes in all, so a router restart or short power cut is ridden out). If all 5 fail, it records the reason in the **system log** - for example "network not found" or "authentication failed - wrong password?" - and restarts in **configuration mode** with its

own access point, so the settings can be checked. The same happens if an established connection drops and can't be restored. Configuration mode stays on until it is left with the system button or **Save and reboot**.

IP address settings:

- **Automatic (DHCP)** Network-assigned IP address (default).
- **Static** Manually set IP address, network mask, gateway and DNS server.

8. Sockets

interfaceNode runs a TCP bridge server that multiple clients can connect to over the network, in either Access Point or Station mode. None of the network services in this section run in **No WiFi** mode.

- **TCP port** Listening port. Must be between 49152 and 65535.

Every TCP client is bridged the same way as USB and RS485 - see [Operation flow](#). Data received from one TCP client is forwarded to USB, RS485, WebSocket clients and every *other* TCP client, never back to the client that sent it. Up to 5 TCP clients can be connected at once. A further client is disconnected as soon as it connects.

The device also advertises itself by mDNS, so compatible software can find it on the network. Only services that have started are advertised:

Service	mDNS type	Port	TXT records (besides device details)
Web interface (configuration mode)	_https._tcp (HTTPS on) and _http._tcp	443 / 80	-
TCP bridge	_dynet._tcp	TCP port	tls (1/0), auth (1/0)
WebSocket bridge	_dynet-ws._tcp	8080	path=/ws , authpath=/auth , tls (1/0), auth (1/0)

TCP bridge security

With TLS on (the default - see [Encryption \(TLS\)](#)), a client must open a TLS session on the TCP port before anything else. The device presents its [certificate](#); clients should check its fingerprint, or trust an uploaded certificate's authority.

Unless TCP access is set to **Open** (see [Security](#)), the first thing a client sends (inside TLS, when TLS is on) must be one line:

```
AUTH <username>:<password>
```

ending with a line feed (`\n`, optionally preceded by `\r`). The device answers:

- **OK** - logged in. The connection then carries DyNet exactly as before; any bytes sent after the line in the same packet are treated as DyNet data.
- **ERR login failed** - wrong username or password, or the first line wasn't an **AUTH** line. The device disconnects.
- **ERR locked <seconds>** - after 5 wrong attempts in a row, logins are refused for 30 seconds, doubling up to 10 minutes. The device disconnects.

Each answer ends with `\r\n`. Until it has logged in, a client receives nothing. A client that hasn't finished the TLS handshake and logged in within 10 seconds is disconnected.

The TCP port always accepts the network connection itself, whichever mode is set; the device then drops a client using the wrong form straight away and says why in the [system log](#):

- TLS on, plain client: disconnected as soon as it sends anything that isn't a TLS handshake, or after 3 seconds if it sends nothing.
- TLS off, TLS client: disconnected as soon as its TLS handshake arrives. With **Open** access a client is bridged as soon as it connects (after the TLS handshake, if TLS is on).

Note: most existing DyNet software that connects over plain TCP can't use TLS or send a login line. For those, turn TLS off and choose **Open** for the TCP bridge - only on a network you trust.

WebSocket bridge

interfaceNode also runs a dedicated WebSocket server at `wss://<device address>:8080/ws` (`ws://` with TLS off - see [Encryption \(TLS\)](#)), separate from the TCP bridge and from the configuration web interface. It runs in operating mode only. Up to 5 WebSocket clients can connect at once, and each is bridged the same way as USB, RS485 and TCP. While 5 are connected, a further client's connection attempt times out.

Unless WebSocket access is set to **Open** (see [Security](#)), a client must log in, in the same way as the Signify PDDEG-S gateway (but the PDDEG-S uses `/api/v1/dynet/auth`; in `dynet_ui` set the Auth URL to `/auth`):

1. Request a token: `GET https://<device address>:8080/auth` (`http://` with TLS off) with an **Authorization: Basic** header holding the WebSocket username and password. The reply is a one-time token (32 hex characters), valid for 30 seconds. A wrong username or password gives **401**; after 5 wrong attempts in a row requests are refused (**429**) for 30 seconds, doubling up to 10 minutes.
2. Connect to `wss://<device address>:8080/ws` (`ws://` with TLS off) and send the token as the first message, as text.

Note: with TLS on, a browser page (such as `dynet_ui`) can only connect once the browser accepts the device's self-signed certificate. Open `https://<device address>:8080/auth` in the same browser once and continue past the warning, or upload a certificate the browser trusts (see [Device certificate](#)).

Until it has logged in, a client receives nothing, anything else it sends disconnects it, and it is disconnected after 10 seconds.

A client using the wrong form is refused at once: `ws://` or `http://` while TLS is on fails its TLS handshake, and `wss://` or `https://` while TLS is off is answered **400 Bad Request**. A request for any path other than `/auth` and `/ws` is answered **404**. Each of these is recorded in the [system log](#) with the client's address and the likely cause. With **Open** access a token is not needed; clients configured to fetch one still work. The auth URL allows requests from web pages on other sites (CORS) unless **Web pages from other sites** is turned off under [Security](#).

Data is exchanged as binary WebSocket messages. Clients should send binary messages: the device forwards the bytes of any message it receives, so a text message would put its character codes on the bus rather than the intended bytes.

WebSocket DyNet filtering

With **Protocol awareness** set to **DyNet**, the DyNet frames sent to WebSocket clients can be limited, for example to reduce traffic to a browser-based display. USB, RS485 and TCP are not affected.

Two rules apply to each DyNet type (DyNet 1 and DyNet 2):

1. **Forwarding** - whether that frame type is sent to WebSocket clients at all. By default DyNet 1 is forwarded and DyNet 2 is not. These two switches are not yet in the web interface.
2. **Opcode list** - which opcodes of that type are sent. Only frames whose opcode is listed are forwarded; all others are dropped. The opcode is byte 3 of a DyNet 1 frame and byte 2 of a DyNet 2 frame.

Opcode lists are set in the web interface under **Protocol**. A list is one or more values from **0** to **255**, separated by commas, with a dash for a range:

List	Opcodes sent
0-255	All
1-255	All except opcode 0
1, 5, 20-30	1, 5, and 20 to 30

The defaults are **0-255** for DyNet 1 and **1-2** for DyNet 2. An invalid list is rejected when saving, with a message saying what is wrong.

Data that is not a DyNet frame, and all data when **Protocol awareness** is **None**, is always sent to WebSocket clients.

DyNet discovery beacon

When **Protocol awareness** is set to **DyNet** and the device is operating (not in configuration mode), **interfaceNode** listens for DyNet discovery broadcasts on UDP port **9998** and replies with its MAC address and TCP bridge port.

Note: this is a separate listener - it does not go through the bridge, and discovery traffic is never forwarded to or from any bridge interface. See [Operation flow](#).

9. Serial

- RS-485 baud rate 9600, 19200, 38400, 57600 or 115200.

RS485 duplex mode (full or half) and termination are set with hardware jumpers - see [Connections](#). RS485 is bridged to USB, TCP and WebSocket clients the same way as any other interface - see [Operation flow](#).

Note: when DyNet is enabled, the USB port answers System Builder's connection check. System Builder sends a fixed DyNet 2 frame when it starts to confirm it is connected to an RS485 network; a plain RS232/RS485 converter returns it because its receiver hears its own transmissions. **interfaceNode** sends that one frame straight back to the PC in the same way. It is not forwarded to RS485, TCP or WebSocket clients, and no other USB data is echoed.

10. Led status

Rebooting (all LEDs, momentarily, immediately before the device restarts):

- **White** Rebooting - shown whatever triggered the restart (saving settings, factory reset, a firmware update, or the system button). After a factory reset the LEDs stay white for up to about a minute while the device erases its data.

Status LED:

- **Green** Starting up.
- **Yellow** Configuration mode - stays until the device is rebooted out of it.
- **Magenta** Operating in Access point mode.
- **Blue** Operating in Station mode.
- **Cyan** Operating with WiFi switched off (**No WiFi**). The WiFi LED stays off.

RS485 LED

- **Slow red pulse** Idle - alternates between bright and dim red every 2 seconds.
- **Fast red flicker** RS485 data activity - returns to idle 0.8 seconds after the last data.

WiFi LED

- **Slow blue pulse** No TCP client connected.
- **Solid dim blue** At least one TCP client connected, idle.
- **Fast blue flicker** TCP data activity.

Note: the WiFi LED shows TCP connections and data only. The WiFi mode is shown on the status LED.

11. System

Important: the web configuration interface is only available in configuration mode. In normal operation (the bridge running), it is not available, and the bridge doesn't run in configuration mode. Switching between the two is a reboot - see [System button](#).

- **Save and reboot** When any setting has been changed, a bar appears at the bottom of the page with this button. It stores every setting and reboots into operating mode. If the network settings changed, reconnect to the device at its new address afterwards.
- **Reboot** Restarts the device without changing any settings. Unsaved changes are lost.
- **Factory reset** Erases all user data and reboots into [default mode](#) - see [Factory reset](#).

Each action asks for confirmation before it runs. The browser also warns before leaving the page with unsaved changes.

Factory reset

A factory reset is started by holding the system button for 2 seconds, or with **Factory reset** in the web interface. Both do the same thing: the device restarts, completely erases all user data, then starts in [default mode](#).

Erased:

- All settings, including WiFi network names and passwords, Enterprise (EAP) sign-in details and the DyNet filters
- Uploaded RADIUS/CA certificates
- The device's TLS certificate and private key (a new certificate and key are made on the next start - browsers will warn again)
- All files on the [USB drive](#) (it is reformatted the next time configuration mode starts)
- WiFi data stored by the WiFi driver, and any stored crash report
- The [system log](#)

Not changed: the installed firmware and the device's [default password](#).

Note: erasing takes up to about a minute, during which all LEDs stay white. Keep the device powered until it has started again. If power is lost during the erase, repeat the factory reset.

System log

Recent log output is viewable at </systemlog.html>, colour-coded by severity (error, warning, info, debug, verbose). Like the rest of the web interface, it is only available in configuration mode.

The log is kept in a fixed-size, 128 KB in-memory buffer. When full, it clears and starts again.

The log survives a reboot triggered by the device itself ([Save and reboot](#), [Reboot](#), or a short press of the system button) and continues where it left off. A factory reset clears it. It is cleared by a power cycle or a firmware update, which are logged as "STARTING COLD BOOT"; any other restart logs "STARTING WARM BOOT" and keeps the existing entries.

Security events

Security events are marked [\[security\]](#) in the system log:

Event	Recorded
Login (web interface, WebSocket, TCP)	Success or failure, the client's address, and for a failed login the username typed
Lockout	The interface and how long logins are locked after repeated failures
Rejected WebSocket clients	An invalid or expired login token, or data sent before logging in
Settings saved	Which groups of settings changed, and any security switch (TLS, login, CORS) turned on or off
Device certificate	A new self-signed certificate made, or a certificate uploaded; a RADIUS/CA certificate uploaded
Firmware update	Started (from which address), rejected, or installed (old → new version)
Firmware changed	On the first start of new firmware: the previous and new version
Factory reset	Requested (button or web interface), and carried out
Start-up	Why the device started: power-on, a restart by the firmware, the watchdog, a crash or a brown-out

Passwords, keys, login tokens and setting values are never recorded.

The log stays on the device: it is kept in memory only and can be read only from </systemlog.html>, after logging in, in configuration mode. Nothing is sent anywhere.

To stop recording security events, turn off [Security event log](#) under [Security](#) and save. Turning it off is itself recorded; after that, security events are only written to the debug serial console. Errors and other device messages are still recorded.

Watchdog

The tasks that move data (RS-485 receive and transmit, USB receive and transmit, the TCP bridge and the DyNet beacon) are watched. If one of them stops responding for 10 seconds, the device restarts automatically and records [restarted by the watchdog](#) in the system log.

12. Firmware

- **Firmware file (.bin)** Choose the firmware file.
- **Install firmware** Confirm, then upload the firmware to **interfaceNode**.

A progress bar shows the update as reported by the device itself:

Message	Meaning
Preparing flash...	Making space for the new firmware (a few seconds)
Writing firmware: N%	Firmware received and written so far
Verifying firmware...	Checking the complete firmware
Update complete...	The firmware was accepted and the device is restarting

Do not disconnect or power off the device until the update is complete. If the update fails, the page says so and the device keeps its current firmware. Only one update can run at a time.

Important: with the current firmware, all settings, including WiFi, are restored to their defaults after a firmware update. The device restarts in **default mode**, so note your settings before updating and re-enter them afterwards.

If the new firmware fails to start, the device automatically returns to the previous firmware.

Firmware files @ <https://github.com/AtkinEngineering/interfaceNode/releases>

New releases and security fixes

interfaceNode never connects to the internet, so it can't check for or announce new firmware itself. New firmware, including security fixes, is published on the **Releases** tab of the firmware repository: <https://github.com/AtkinEngineering/interfaceNode/releases>. Check it regularly, or use GitHub's **Watch** → **Custom** → **Releases** on that repository to be notified when a new release is published. The installed version is shown under **Device (Version)**.

Version numbers are **major.minor.patch**:

- **Patch** (for example 1.2.3): security fixes and bug fixes only, no new features. Security fixes are released this way as soon as they are ready, without waiting for the next feature release, and are free of charge.
- **Minor** and **major**: new features or changes.

Support period: Security updates are provided free of charge for 7 years after the product's end of life.

To report a security vulnerability, see **SECURITY.md** in the firmware repository. Reports are acknowledged within 1 week, assessed within 3 weeks and fixed within 8 weeks - within 3 weeks if the vulnerability is being actively exploited.

Each release lists its changes in the changelog, and security fixes are marked as such. Every release package also includes a software bill of materials (**SBOM.spdx**) listing the software components in the firmware, and a report (**SBOM_report.md**) of known vulnerabilities checked against them at release time.

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[^usbpower]: USB Type-C port does not supply power.

[^rs485power]: RS-485 (parallel RJ12 6P6 ports) supplies power & data to **interfaceNode**.

[^jumpers]: Termination and duplex are both set via jumpers internal to the enclosure, not configurable from the web interface - changing either requires opening the unit. For duplex specifically: the jumper selects between the receiver always being enabled (full duplex) or tied to the transmit-enable line (half duplex). The firmware drives a single direction-control signal in either case - the jumper, not the firmware, determines which duplex mode results.

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